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Adaptive virtual agent: Design and evaluation for real-time human-agent interaction

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ABSTRACT

When we converse, we adapt our behaviors to our interlocutors. The adaptation can serve to indicate our engagement which can also elicit enhancement of the involvement of others. Virtual agents (or socially interactive virtual agents) that play the role of interaction partners can improve the human users' interaction experience by displaying continuous and adaptive behaviors in real time. Virtual agents have been used in multiple domains to improve user interaction and performance. The promising results of the endowment of adaptation to agents in increasing the agents' perception and user experience were shown in previous studies. In this paper, we develop an adaptive virtual agent that renders real-time adaptive behaviors based on the behaviors shown by its human interlocutor. The ASAP model rendering reciprocally adaptive agent behavior was employed to realize the system. The system consists of four main parts: perception of social signals, agent adaptive behavior generation, agent visualization (i.e. rendering of the agent's verbal and nonverbal behavior), and communication of signals. To showcase the usefulness of our adaptive agent, as a proof-of-concept we choose the e-health application of cognitive behavior therapy (CBT), which identifies and rectifies biased and irrational thoughts (or automatic thoughts). Through this study, we show the importance of giving the agent reciprocal adaptation capability notably in enhancing the user experience and the effectiveness of the CBT session. We validate the importance of endowing such adaptation capability by studying the difference between agents that are reciprocally adaptive, solely expressive (with mismatched behavior), and inexpressive (in a still posture) via questionnaires and measures related to the agent perception (naturalness, human-likeliness, synchrony, and engagement) for user experience and the CBT effectiveness (mood, anxiety, stress, and cognitive change). These results highlight the value of making virtual agents adapt in real time. This could lead to agents being capable of providing more personalized and interactive experiences for a wide range of applications. Also, we have collected a new human-agent interaction (HAI) database, HAI-CBT database, which is publicly available to the research community.

1. Introduction

We communicate by expressing our thoughts and feelings through the exchange of verbal and nonverbal signals. The transfer of social signals is not only dependent on the conveying content but also on the interacting partners as we adapt to them (Burgoon et al., 1995). Adaptation is a key aspect of interpersonal relationships (Cappella, 1991). It can serve to indicate our engagement and rapport which can also elicit enhancement of involvement of others (Delaherche et al., 2012; Gupta et al., 2019; Huang et al., 2010; Raffard et al., 2018). Interlocutors adapt their behaviors throughout the interaction continuously, reciprocally, and dynamically to those of the others, referred to as reciprocal adaptation (Woo et al., 2023d). The reciprocal adaptation arises between interlocutors in real time following a looped process.

Strengthening interpersonal relationships is important in any task in which people work together. In particular, in psychotherapy, including cognitive behavior therapy (CBT) (Beck, 2020), the development of rapport (VandenBos, 2007) and the collaborative relationship between the supporter and the help-seeker has been emphasized, and their impact on treatment effectiveness has been investigated. Relationships of mutual understanding, acceptance, and sympathetic compatibility between or among individuals have been shown to contribute to the effectiveness of psychotherapy (DeVault et al., 2014; Huang et al., 2010;

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Raffard et al., 2018). Adaptation, both verbal and nonverbal, strengthens the relationship between supporter and help-seeker, resulting in help-seekers feeling more at ease and more likely to confront their problematic relationships and improving adherence and persistence rate with the supporter's suggestions. In medical and psychological fields, it is known that rapport affects the effectiveness of CBT (Asay and Lambert, 1999; Ardito and Rabellino, 2011; Norcross and Lambert, 2018). During real therapy between a patient and a therapist, health support is provided through face-to-face interactions. The therapist not only provides the therapy through verbal communication but also expresses the feeling of sympathy and engagement nonverbally with their patient (Ramseyer and Tschacher, 2014; Koole and Tschacher, 2016).

Virtual agents interact with human users by playing the role of interlocutor. Their central objective is to improve the human users' interaction experience by increasing their users' engagement level. A way to attain their goal is to adapt their behaviors depending on those of their users. For such embodied agents, they need to display continuous and adaptive behaviors in real time. Adaptive agents, adapting their verbal and/or nonverbal behavior, have demonstrated their use in increasing the user engagement (Schroder et al., 2011; Ritschel et al., 2017; Weber et al., 2018; Biancardi et al., 2021; Woo et al., 2023d), rapport (Huang et al., 2010; Raffard et al., 2018), interaction synchrony (Raffard et al., 2018; Woo et al., 2023d), and impression of the agent (liking, naturalness, and human-likeliness) (Bailenson and Yee, 2005; Huang et al., 2010; Biancardi et al., 2021; Woo et al., 2023d). The real-time aspect of behavior generation along with the fluid dialog management needs to be assured throughout the whole interaction for both interlocutor roles of a listener and a speaker which is not a trivial task.

The use of virtual agents can be seen in various domains ranging from assistance (Sidner et al., 2018; Biancardi et al., 2021) to healthcare (Philip et al., 2020; Bickmore, 2022; Shidara et al., 2022). Virtual agents have been demonstrated to be promising tools, notably for medical care, in gaining users' trust and acceptance (Philip et al., 2020). Several studies have highlighted the benefits of using virtual agents in e-health applications (Philip et al., 2020; Lisetti et al., 2013; Lucas et al., 2014; Bickmore, 2022). As such several conversational agents have been developed to deliver CBT focusing on the treatment (Ring et al., 2016; Fitzpatrick et al., 2017; Kimani et al., 2019; Shidara et al., 2022). However, human supporters communicate with their help-seekers through behavior (both verbal and nonverbal) that are adapted to that of their help-seekers (Ramseyer and Tschacher, 2008). It is thus important for virtual CBT agents to also communicate verbally and nonverbally, and adapt their behavior to that of their users. It is not clear from previous studies whether behaviors with reciprocal adaptation, more specifically facial expressions and head movements, have an impact on improving the effectiveness of human-agent interaction during CBT.

The contributions of this study are twofold: (1) to elucidate whether adaptable virtual agents can enhance the experience perceived by users themselves, and (2) to ascertain whether adaptable virtual agents can improve the effectiveness of CBT through comparative experiments. We develop an adaptive virtual agent that renders adaptive behavior in real time based on the behavior shown by human interlocutors. To generate the agent's behavior in our system, we adopted the ASAP model, a computational model for rendering mutually adaptive agent behavior (Woo et al., 2023d). Our system loops through the processes of social signal perception, agent adaptive behavior generation, agent visualization, and signal transmission, ensuring real-time responsiveness. Furthermore, we demonstrate that nonverbal adaptation of virtual agents contributes to the achievement of interaction objectives in pairs of users and virtual agents. We check that the endowment of adaptation capacity further enables agents to provide a better user experience than a solely expressive agent. For this proof-of-concept, we incorporate a scenario based on CBT (Beck, 1979, 2020) into the virtual agent. CBT is an established mental healthcare method that provides face-to-face dialogs with users, similar to our system's setup. As CBT is effective not only for mental illnesses such as depression and anxiety disorders but also for coping with daily psychological distress, it becomes a suitable option for the general public (Greenberger and Padesky, 2015). In this study, we target the general population and conduct interactions based on CBT to cope with daily psychological distress. This research reveals how the adaptation of nonverbal behavior affects the relationship that users feel with the agent and how it impacts the objectives of the interaction.

Our proposed method of endowing reciprocal adaptation capability to virtual agents is indeed applicable to other therapies. In CBT, the expected effects of reciprocal adaptation on the agent-user relationship correspond to general therapeutic skills (Young and Beck, 1980; Wright et al., 2017). Therefore, while not limited to CBT, this method is also anticipated to be effective in other therapies. However, we specifically evaluated its application in CBT for two main reasons (Beck, 1979). First, CBT is one of the primary mental healthcare approaches for dealing with negative moods and stress. Additionally, CBT is an effective psychotherapy for mood regulation in daily life, reaching a broad audience. The second reason for focusing on CBT is its specific goal of addressing negative automatic thoughts and mood improvement within a session. This focus enables appropriate observation of changes through questionnaires, even before and after a single session. Consequently, this facilitates a deeper exploration of the effects of reciprocal adaptation in conjunction with CBT's outcomes.

We precisely look into the nonverbal behavior of the head (facial expression and head/gaze rotation) studying the difference between agents that are reciprocally adaptive, solely expressive (with mismatched behavior), and inexpressive (in a still posture). Additionally, we analyze how the relationship with the agent and the goals of the interaction are interrelated. We measure the user experience via the reported agent perception (naturalness, human-likeliness, synchrony, and engagement) and the CBT effectiveness (mood, anxiety, stress, and cognitive change). Via this study, we have also collected a new human-agent interaction (HAI) database called HAI-CBT database[1] consisting of 60 human-agent interaction CBT sessions.

The rest of the paper is structured as the following: Section 2 presents state-of-the-art for reciprocal adaptation in human-agent interaction and virtual agents applied for mental health care; Section 3 introduces our study method consisting of the description of our real-time expressive and adaptive agent system, our proof-of-concept (CBT scenario and its integration), and our experimental design; Section 4 provides objective and subjective evaluation results of the user experience about the agent's behavior perception and the CBT effectiveness via the user's mood and state change; Section 5 discusses our results along with the limitation and future work of our study; and Section 6 summarizes our findings.

2. Related work

This section outlines relevant research on how the reciprocal adaptation of virtual agents has been modeled and can influence user perceptions in communication and the impact of reciprocal adaptation on mental health care, which is key to our interest.

2.1. Reciprocal adaptation in human agent interaction

Adapting to interlocutors is an essential part of interaction. Agents that interact with human users by taking the role of an interlocutor should also have the skill of adaptation. Related research has worked on creating conversational agents (virtual agents, humanoid robots,

[1] The HAI-CBT database is available on request after signing the EULA form: https://cloud.isir.upmc.fr/s/k7yToEQTQ7rHdRj.

chatbots) that can adapt to their users. The adaptation can be done at different levels shown via various social cues (verbal, nonverbal, and/or conversational strategy) which are employed for diverse applications such as providing information, company, assistance, education, and medical care.

For the interaction to be personalized based on the interlocutor, several works have focused on adapting the agent's verbal context and/or conversational strategy. Nishimura et al. (2007) created a dialog system for chatbots that generates natural responses along with the response timing based on the user's inputs via decision trees. Ritschel et al. (2017) looked at the variation of linguistic style and its impact. The linguistic style was used to represent the robot's personality. The adaptation of linguistic style was modeled by a reinforcement learning model based on the user's engagement level which was estimated from the user's gaze and posture. Their study showed that adapting linguistic style can improve the user's engagement. Weber et al. (2018) studied the adaptation of jokes based on the user's sense of humor. The user's humor was detected without explicit user feedback through the user's smile and laughter. They proposed a robot that performs real-time adaptation using reinforcement learning and demonstrated that their robot performs significantly better in terms of amusement level by making jokes that consider its user's sense of humor compared to those that produce jokes randomly. Ding et al. (2022) created a conversational agent, TalkTive, that generates backchannels aiming to help the elderly to be engaged during cognitive assessments. TalkTive predicts the verbal backchanneling form that can be either reactive backchannels (e.g. "hmm") or proactive backchannels (e.g. "please keep going") and its timing.

The nonverbal channel plays a major role in communication. Other works have concentrated on modeling the nonverbal adaptation of agents. Huang et al. (2010) designed a virtual agent that produces visual backchannels via conditional random fields (CRFs) from the user's gaze, prosody, and lexical features. Their study denoted that visually adapted backchannels can reinforce the rapport agents build with their users and can be perceived as more natural. Schroder et al. (2011) also developed a virtual agent that produces nonverbal backchannels (smile, head movement, and vocalization). The agent displaying backchannels was able to engage its user better than an agent that does not. Agent's adaptation has also been expressed through the production of mimicry behavior. Bailenson and Yee (2005) created a virtual agent that renders the mimicry behavior. It imitates the user's head movements within a delay of 4 s. A mimicking agent was shown to be perceived as more positive and persuasive than one that does not mimic the user. Raffard et al. (2018) also assessed the effect of mimicking virtual agents, with a mimicry delay varying between 0.5 s and 4 s, with participants suffering from schizophrenia and healthy participants. Their results revealed that the rapport and interaction synchrony was improved with the display of mimicry behavior for both participant groups. Adaptive nonverbal behavior is not only the one that is produced as a reaction to the user or copying the user's behavior but also the one that matches the user's behavior. Anderson et al. (2013) made a virtual agent framework for job interviews facilitating self-reflection and providing personalized coaching. The agent serves as a virtual recruiter that generates its nonverbal behaviors according to the user's face and hand gestures. Pereira Santos et al. (2023) built an embodied agent for obstetric simulation training. The agent playing the role of a digital patient adapts its facial expressions in real time via the behavior that is commanded by an on-screen controller. Sidner et al. (2018) realized a real-time architecture for companion agents (virtual agents and robots) that provides companionship for the elderly. The agent adapts its gesture via the user's facial gestures and motions. Biancardi et al. (2021) created a virtual agent that adapts its behaviors to that of its interlocutor. It serves as a virtual museum guide and aims to maximize the user's engagement. Their virtual agent is capable of adapting its nonverbal cues at the behavioral and conversational levels. They demonstrated that the adaptive agent was perceived as more positive than the non-adaptive agent. Amini et al. (2021) proposed an agent for health interventions that adapts its social behavior based on its user's behavior. They modeled rapport using two separate models for speaking and listening modes. For each sentence uttered or listened to (both agent and user), the user's behavior is recognized and the agent's nonverbal behavior is returned. Their results showed that such adaptive agents improved the user experience for health interventions in terms of user's attitude, intention to (re-)use the agent, perceived enjoyment, perceived sociability, perceived usefulness, social presence, and trust. Recently, Woo et al. (2023d) built the ASAP model which endows reciprocal adaptation capacity to the virtual agent. The virtual agent is capable of mutually adapting its behavior (facial expression and head/gaze movement) to that of its human user. Their study conducted a third-party evaluation of the human-agent interaction video clips and showed that their virtual agents are perceived as natural, human-like, in sync, and engaged with their human interlocutors.

2.2. Mental health care with virtual agents

CBT is an effective and well-established healthcare method for addressing mental illnesses such as depression and anxiety and for daily health care. Although CBT is effective, its lack of accessibility is a serious issue, as it requires a high level of skill for supporters. Various types of conversational agents have been employed to promote the use of CBT. Conversational agents include text-based agents, such as those used in messaging applications, robotic platforms, and virtual agents. Among them, text-based agents are particularly common due to their ease of use and are often used as smartphone applications for convenient access. Two examples of such applications are Wysa (Inkster et al., 2018) and Woebot (Fitzpatrick et al., 2017), which have been tested for their ability to assist individuals exhibiting mild to moderate depression and anxiety symptoms. These tools are designed to provide regular mental health care interventions rather than clinical treatment. On the other hand, one limitation of these text-based agents is that they communicate only via language. In mental health care, interactions using nonverbal behavioral modalities are important for improving the relationship between the supporter and the help-seeker, as they influence various aspects, such as understanding the psychological state of the other person, empathic behavior, and sense of presence.

Virtual agents and robots have the advantage of face-to-face multimodal interaction, including facial expressions, gestures, and voice. In particular, virtual agents can be used at a relatively low cost, and their appearance and voice can be customized. For these reasons, virtual agents are anticipated for use in mental health care. DeVault et al. (2014) conducted experiments of medical interviews with virtual agents that operated using the Wizard-of-OZ method or autonomously. The results showed that users felt a rapport with the agent. It was also suggested that an autonomous agent without an operator behind them could facilitate user self-disclosure. In addition, attempts are being made to use virtual agents not only for medical interviews for screening purposes but also for mental health care, such as CBT. Shidara et al. (2022) developed a virtual agent that delivers CBT which helps users to identify and rectify automatic thoughts. They alternated the conversational strategy for participants that needed help in identifying their automatic thought via a language model-based automatic thought classifier. Ring et al. (2016) also illustrated a virtual therapist agent for CBT counseling. The system dialog is managed with the user's speech and affect (detected from the user's speech). Along with the dialog management, they generate the agent's nonverbal behavior, which is automated via the Behavior Expression Animation Toolkit (BEAT) (Cassell et al., 2001). The potential efficacy of affect-aware agents for the guidance of CBT scenarios is presented. Efforts have also been made to improve user receptivity by adjusting the appearance of the agent, its set age, etc. Parmar et al. (2022) have systematically manipulated animation quality, speech quality, rendering style, and

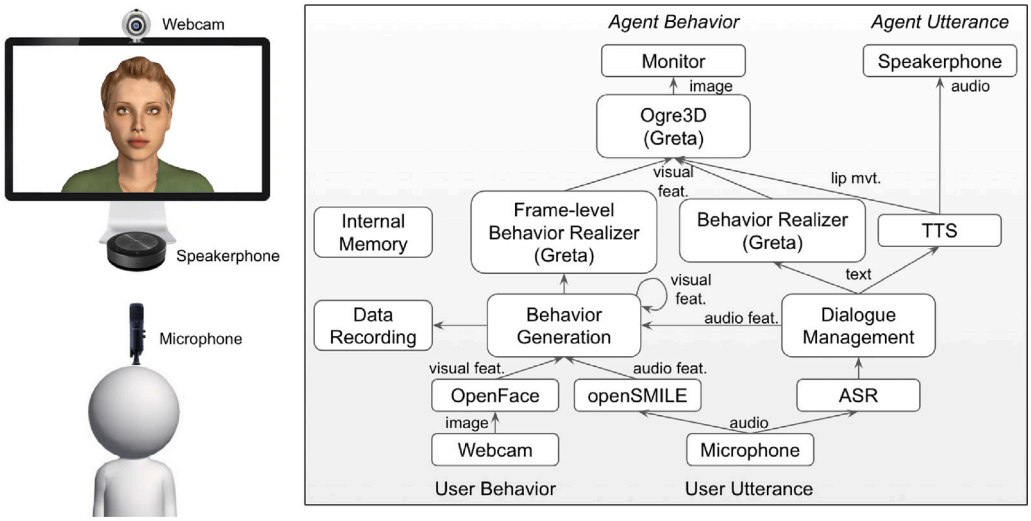


Fig. 1. Real-time expressive and adaptive agent system setup (left) and architecture (right). The proposed system interacts with the user via a virtual agent that shows expressive and adaptive behavior in real time. It captures the user's face with a webcam and the user's speech with a microphone. The agent is displayed in front of the user on a monitor and its speech is rendered via a speech synthesizer and a speakerphone.

simulated empathy in the domain of health counseling. They investigate the effects on virtual agents' perceptions in terms of spontaneity, engagement, trust, credibility, and persuasion. The results showed that the agents improved their ability to persuade but hindered their ability to improve trust. In terms of agent design, suggestions include agents tailored to black church communities (O'Leary et al., 2020) and agents with the appearance of older adults (Razavi et al., 2022) to provide realistic conversational practice to older adults at risk for isolation and social anxiety. While adaptations such as these have been made, the mental health impact of real-time adjustment of nonverbal behaviors such as facial expressions and head movements is not clear. Boustani et al. (2021) developed a digital health agent delivering online motivational interviewing for alcohol misuse displaying empathic rule-based verbal and nonverbal behaviors. Their participants reported positive experiences with the agent including engagement, acceptance, utility, and intent to use the agent.

In this study, we seek to develop a virtual agent that is capable of adapting its behavior to its interacting user. We focus mainly on the nonverbal adaptation of the agent generating expressive and adaptive agent's facial expressions and head/gaze movements. We integrate the ASAP model (Woo et al., 2023d) to enable the reciprocal adaptation to virtual agents. We intend to check how human interactants perceive adaptive agents in terms of naturalness, human-likeliness, interaction synchrony, and engagement. We aim to also adapt the conversational strategy (or conversational move) for our selected scenario, the CBT scenario, to the user's response. Compared to the previous works, we tried to assure the real-time functioning of our agent at the frame-level to display continuous agent movements. Moreover, to our knowledge, no nonverbal adaptive agent has yet been introduced for the application of CBT. We intend to improve the interaction itself, more precisely the user experience (perception of the agent), and at the same time the effect of CBT compared to previous CBT systems (Ring et al., 2016; Shidara et al., 2022) by rendering adaptive agent behavior.

3. Method

To improve the user's interaction experience and the perception of the agent, we built a real-time expressive and adaptive virtual system using the ASAP model, to render reciprocally adaptive agent behaviors. Our system, illustrated in Fig. 1, is composed of 4 main functionalities: perception of the agent's own behavior (via OpenFace, openSMILE, and ASR), generation of expressive and adaptive behavior (via behavior generation module), dialog management (via dialog management module), and visualization of the agent's behavior (via frame-level behavior realizer and behavior realizer modules and Ogre3D). For the effectiveness assessment of our system, CBT is chosen as our proof-of-concept to validate our experimental hypotheses stated in the upcoming sections.

3.1. Real-time expressive and adaptive agent system

For an agent to be capable of interacting in real time and displaying expressive and adaptive behavior, it needs to possess the following functionalities:

- perception of the user's behavior;
- perception of its own behavior;
- generation of expressive and adaptive behavior;
- dialog management;
- visualization of its behavior.

Each of the functionalities is endowed to our agent[2] by employing the corresponding techniques.

3.1.1. Perception of user's behavior

An agent interacting with a human user needs to act accordingly to its user. It needs to take into account the user's behavior (speech and gesture). To do so, the agent first needs to perceive such signals. The perception can be separated into three parts:

- User's speech content;
- User's speech prosody;
- User's gesture.

User's speech content. The user's speech content is important as the user's intentions are explicitly expressed via words. This information is essential for all types of automatic systems interacting with the user which is easily seen in conversational AI assistants such as Google, Alexa, and Siri. To capture this information of speech content, we choose to integrate the Google ASR.[3] Automatic Speech Recognition (ASR), also referred to as Speech-to-Text (STT), is the transcription technology that captures the audio of spoken words and transforms it into written text. The ASR technique identifies full phrases and transcribes them as the user is speaking.

[2] The code is available here: https://github.com/isir/greta/wiki/IAVA-for-Adaptive-Behavior-Generation.

[3] https://cloud.google.com/speech-to-text

User's speech prosody. When we speak, our intentions are conveyed via the speech content but we also vary the way we speak using our voice qualities. Our speech can vary in pitch (high or low), loudness (loud or soft), and duration (fast or slow). This variation in speech is referred to as speech prosody. We extract the speech prosody using openSMILE (Eyben et al., 2010), an open-source toolkit for audio feature extraction. We choose to retrieve the prosodic features of:

- Fundamental frequency representing the pitch;
- Loudness quantifying the sound energy;
- Voicing probability estimating a percentage of unvoiced and voiced energy;
- Mel-frequency Cepstral Coefficient (MFCC) (Logan, 2000) which is a representation of the short-term power spectrum of a sound.

These features are obtained in real time with a frequency of 100 Hz.

User's gesture. For the gesture, we decide to focus on the perception and the generation of the facial gestures (facial expressions and head/gaze movements) of both the human user and the agent. For the perception of the user's facial gestures, we use OpenFace (Baltrušaitis et al., 2016), an open-source toolkit, to extract the facial features at the frame-rate of 30 fps which are as follows.

- Gaze movements (G_x and G_y) which are the gaze angles w.r.t. the x and y axis;
- Head movements (R_x, R_y, and R_z) which are the Euler head rotations w.r.t. the x, y, and z axis;
- Facial expressions via facial Action Units (AUs) (Ekman and Friesen, 1976) which are facial muscle movements defined by the Facial Action Coding Systems (FACS) (Ekman and Friesen, 1978). We use *AU1* (inner brow raiser), *AU2* (outer brow raiser), *AU4* (brow lowerer), *AU5* (upper lid raiser), *AU6* (cheek raiser), and *AU7* (lid tightened) which are the most representative muscle movements for facial expression that are not related to speech.

3.1.2. Perception of agent's behavior

The agent needs to behave not only depending on its user's behavior but also with respect to its previously displayed behavior and its current intention (e.g. lip movement matching its speech). For the agent to act as such, it needs to also perceive its own behavior. For this, the agent remembers its previous behaviors (speech content and prosody computed in real time via openSMILE at 100 Hz, and previously rendered facial gestures) within its internal memory, which keeps the 100 most recently seen user behavior and displayed agent behavior, along with its currently exhibiting behavior.

3.1.3. Generation of expressive and adaptive agent's behavior

The behavior signals that are perceived by the human user and the agent are used to generate expressive and adaptive agent behavior. We employ the Augmented Self-Attention Pruning (ASAP) model (Woo et al., 2023d). This model generates expressive agent facial gestures that are reciprocally adaptive. It learns interpersonal relationships via real human-human interactions, from a corpus of screen-mediated face-to-face interactions, the NoXi database (Cafaro et al., 2017), with its self-attention pruning and data augmentation techniques. It also assures the movement continuity of the generated behavior by performing autoregressive adaptive online prediction. We chose to use the NoXi database as it meets the specific recording setting required to train the ASAP model which is recording interlocutors with cameras facing them forward. No other database met this recording requirement and there were no CBT databases that were publicly available.

The pre-trained ASAP model is integrated into our system. The perceived visual and audio features of the past 100 time-steps of both human user and agent are passed to the ASAP model to render the agent's expressive and adaptive visual behavior at the next time-step. The communication protocols of ZeroMQ[4] (Hintjens, 2013) and OSC (Open Sound Control)[5] (Wright, 2005) are used to pass the signals (for visual and audio signals respectively), after syncing the different sampling rates to 25 Hz which is the ASAP model's sampling rate, to the model for the prediction. The model generates the agent's behavior with an inference speed of 0.008 s.

3.1.4. Dialog management

During a conversation, the dialog is managed between interlocutors by taking turns, and the next flow of discussion is decided depending on the speech content. The agent also needs to be able to behave similarly in a real-time conversation with its human user. To this aim, we use Flipper2.0 (or Flipper) (van Waterschoot et al., 2018) which is a dialog engine that can flexibly direct the conversational flow. The utterance text of the user's speech is obtained from the ASR and passed to Flipper via ActiveMQ[6] (Snyder et al., 2011), a communication protocol, to choose the next conversational move.

3.1.5. Visualization of agent

For the visualization of the animation of the virtual agent, we use the Greta platform (Niewiadomski et al., 2009) which is an open-source virtual agent platform simulating an agent's verbal and nonverbal behavior in real time. The agent's speech (its next conversational move chosen by the Flipper dialog engine) is transformed from text to audio using the CereProc[7] speech synthesizer (or Text-to-Speech (TTS)) within the internal audio module. The matching mouth movements are also produced along with the speech audio by the internal behavior realizer module. For the display of the agent's adaptive facial gesture, generated by the integrated ASAP model, the predictions are passed to the frame-level behavior realizer module, detailed in Woo et al. (2023c). The agent's behavior outputted by the behavior realizer modules (adaptive agent behavior and mouth movements) is then passed to Ogre3D,[8] an open-source scene-oriented 3D rendering engine, for display.

3.1.6. System setup and performance

For the system setup, a virtual agent is displayed in front of the user on a monitor (in a close-up of their face, head, and shoulders), as depicted in Figs. 1 and 2. The user's speech is captured via a microphone and the user's facial gestures (head and gaze movements, and facial expressions) are obtained through a 1080p RGB webcam. The agent's spoken utterance (speech of the chosen conversational move) is rendered using a speech synthesizer and a speakerphone.

The system runs on two computers in parallel. The first computer continuously displays the agent's behavior via the Greta platform. The second computer runs the ASAP model, generating expressive and adaptive agent behavior in real time, along with the perception toolkits of OpenFace and openSMILE (for facial feature extraction and prosodic feature extraction respectively).

For the system's performance, real-time functioning (25 Hz) is assured. A single system loop execution time of 0.04 s with no delay is assured. To detail, the system loop consists of perception (approx. 0.03 s), behavior generation (approx. 0.008 s), communication (approx. 0.001 s), and visualization (approx. 0.001 s) which are all synced. Further information on the system specifications can be found in Woo et al. (2023b).

[4] https://zeromq.org
[5] https://opensoundcontrol.org
[6] https://activemq.apache.org
[7] https://www.cereproc.com
[8] https://www.ogre3d.org/

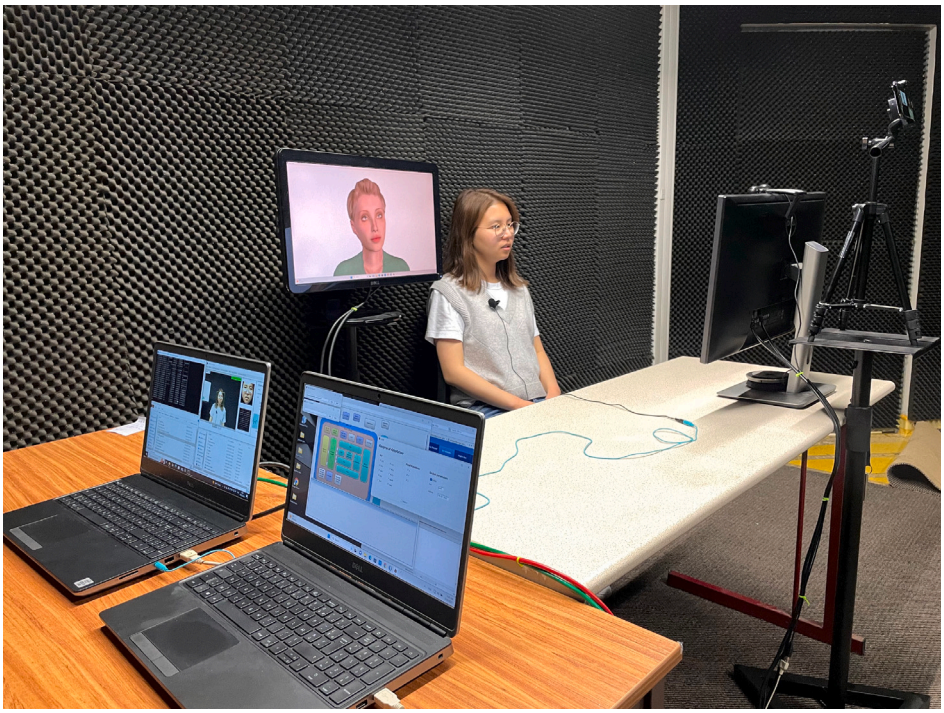


Fig. 2. Experimental setup of the research site.

3.2. Cognitive behavior therapy

To assess if endowing virtual agents with reciprocal adaptation mechanisms enhances their user's experience, we choose to use the CBT scenario as our proof-of-concept.

CBT (Beck, 1979, 2020) is a mental health treatment that restructures automatic thoughts (or irrational thoughts). These automatic thoughts are those that come up to our minds suddenly and unconsciously. Because of their nature of occurring unexpectedly, we are not aware of them but they affect our mood. These thoughts often elicit negative feelings but can also evoke misleading positive emotions. To help people to recognize and rectify automatic thoughts into balanced ones, the CBT treatment is delivered. The restructuring of thoughts is done by asking the participants several fact-finding questions to guide them through the process of identification of such thoughts and changing them. The key effect of CBT is that it clarifies irrational situations and thoughts, and brightens people's moods.

By applying our adaptive virtual agent to CBT, we expect to see an amelioration in the effectiveness of CBT in mood improvement along with the user's experience, notably the enhancement of the agent's perception.

3.2.1. Scenario

We choose to work with the CBT scenario presented in Shidara et al. (2022) which was created with a psychiatrist and is presented in Fig. 3. We follow the same scenario (translated to French) of asking the participants to self-report their mood (negative mood intensities asked before and after the CBT session via questions Q3 and Q14), helping them to identify their automatic thought (via question Q4), and guiding them with fact-finding questions (by asking questions concerning proof, disproof, disputing, Socratic questioning, balanced thought, and caring words). For the identification of participants' automatic thought, their response is verified by an automatic thought classifier model (detailed in Section 3.2.2) to check if it corresponds to an automatic thought. The model performs a classification on the participants' response by checking whether it is based on negatively distorted cognition or factual validity to judge if they have correctly identified an automatic thought. The next conversational move (agent's utterance) is selected depending on whether the user's answer is an automatic thought or not.

Table 1
Classification results of automatic thought for French. D: data collected in Shidara et al. (2022), G: sentences from Greenberger and Padesky (2015), and the score in bold represents the best score.

Training data	Test data	Feature	Accuracy	F1-score
D	D	TF-IDF	0.73	0.81
		BERT	0.67	0.76
		TF-IDF + BERT	0.67	0.76
G	D	TF-IDF	0.70	0.79
		BERT	0.70	0.79
		TF-IDF + BERT	0.67	0.78
D + G	D	TF-IDF	**0.76**	**0.83**
		BERT	0.73	0.79
		TF-IDF + BERT	0.73	0.79

3.2.2. Automatic thought classifier

The automatic thought classifier is a classification model that serves to validate whether the participants have successfully identified their automatic thought. The model is based on the Support Vector Machine (SVM) classifier algorithm with a linear kernel. We get the French word embeddings from a pre-trained language model for word representations, the Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2018), and use the embeddings to train the automatic thought classifier. The choice of using BERT embeddings was chosen after testing different combinations of training data and distributed representations of Term Frequency-Inverse Document Frequency (TF-IDF), BERT, and both TF-IDF and BERT. The classification results for French are reported in Table 1 which shows that TF-IDF has the best accuracy and F1-score of 0.76 and 0.83 respectively when evaluated on sentences from Greenberger and Padesky (2015) and trained with both data collected in Shidara et al. (2022) and sentences from Greenberger and Padesky (2015). However, through supplementary evaluation of French automatic thought sentences (more complex and longer sentences), the classification with TF-IDF failed while the one with BERT (with similar accuracy and F1-score of 0.73 and 0.79 respectively) performed well. Thus, BERT was selected as our input feature for the automatic thought classifier.

No.	Item	Sentence (Virtual agent's questions)
		Hello, my name is Mai, and I'm a therapist. I'm here to help you learn how to face your worries.
Q1	Situation	*Has anything been bothering you lately, maybe something that's difficult to deal with or to face?*
		Please describe anything that is painful or burdensome.
Q2	Mood	*How did your mood change that time?*
Q3	Mood score (0–100%)	*How would you describe your mood's intensity from 0 to 100?*
Q4	Identification of an automatic thought	*I want to know why you felt like that in that situation.* *What thoughts came to you when you faced that event?*
		I see...
Q5	Proof	*Such thoughts that surface when we face certain events are called automatic thoughts.* *Since automatic thoughts are unconscious, they sometimes seem true.* *If your automatic thought is accurate, what do you think it is based on?*
Q6	Disproof	*So, do you have any other thoughts about that situation?* *For example, does anything contradict your automatic thoughts?*
Q7	Disputing(confirmation)	*Do you have any other thoughts? If so, can you share them?*
Q8	Socratic questioning* 1	*Let's examine the basis of the automatic thought that we have considered so far as well as how thoughts are balanced by contrary facts. Let me ask you some questions.*
Q9	Socratic questioning 2	*What is the worst possible consequence of this situation?*
Q10	Socratic questioning 3	*Looking at the other side, what is the end result when you act as you want?* *Please describe the most realistic scenario based on the predictions you just made.*
Q11	Balanced thought	*From that answer, let's create a new thought. How can you rethink this event?*
Q12	Caring words 1	*Good. Do you have anyone who will listen to you talk about this problem?*
Q13	Caring words 2	*I see, can you manage that?*
Q14	Mood score after change (0 to 100%)	*Well, our time is up for today. Thank you for your hard work. Please feel free to call again.* *Please feel free to talk to me. How has the intensity of your original mood changed?* *Express this on a scale from 0 to 100.*
[End]	Concluding a session	*If it's different from the beginning, that suggests that you internalized my suggestions well.* *That's it for today. Thank you for your hard work. Please feel free to call again.*

*Questions that create awareness of inconsistency in thoughts.

Fig. 3. Scenario of utterances spoken by system used in Shidara et al. (2022) under the copyright terms of CC BY.

When inputting the participant's response (user's utterance), the model identifies automatic thoughts by performing binary classification. If the response was identified by the classifier as an automatic thought, the agent moves on to the next item of the scenario. However, if it was determined not to be an automatic thought, the agent provides a hint and asks the user again for an automatic thought. The agent provides at most six hints (from Beck, 2020) and if the user was unsuccessful to answer with an automatic thought even after six attempts, failing for the seventh time, the agent moves on to the next item. The automatic thought classifier is integrated within the dialog management module (cf. Section 3.1.4 and Fig. 1).

3.3. Experiment

3.3.1. Hypotheses

To investigate the usefulness of a virtual agent showing expressive and adaptive behavior, we propose three experimental conditions which are as follows:

- RA: Reciprocal Adaptation (agent showing expressive and adaptive behavior);
- MM: Mismatched Movement (agent displaying pre-registered mismatched behavior that is expressive but not adaptive);
- SP: Still Posture (agent in a still posture with only mouth movements) same condition as in Shidara et al. (2022).

We build our study upon the work of Shidara et al. (2022) which demonstrated that interacting with virtual agents helps to improve the effectiveness of CBT (in improving people's mood and state).

For our study, we hypothesize that adaptive and expressive agents can further enhance the user experience (agent perception in terms of behavior naturalness, human-likeness, synchrony, and engagement) and the effectiveness of CBT (user mood and state change) based on the previous observation that virtual agents with adaptation help improve the interaction. Thus, we expect agents enabled with reciprocal adaptation (RA) to outperform agents showing random behaviors (or mismatched behavior; MM) and still agents (or in a still position; SP).

Our hypotheses are the following:

- H1: Reciprocally adaptive agents are more positively perceived and more effective for CBT compared to agents showing mismatched behaviors (RA > MM);
- H2: Reciprocally adaptive agents are more positively perceived and more effective for CBT compared to agents in a still position (RA > SP).

3.3.2. Protocol of experiment

For the experiment of the interaction of human users with our agent system to proceed smoothly, we conduct our experiment with the following protocol:

(i) Oral explication.
(ii) Signing consent forms.
(iii) Lecture of a document on automatic thought.
(iv) Filling out pre-questionnaires.
(v) Interaction between the participant and our virtual agent.
(vi) Filling out post-questionnaires.

The detail of the protocol is as follows. We start the experiment by giving the participants an oral explication of the aim of the study and the experimental protocol (pre-questionnaire, interaction with our agent, and post-questionnaires) is given to the participant. The participants sign two consent forms: one to participate in the experiment and the other to give the authorization of use concerning the collected data. The participant is then invited to read a document explaining what an automatic thought is for a better understanding of the CBT scenario. They start by filling out the pre-questionnaires before the interaction. After the interaction with our virtual agent, they finish by filling out post-questionnaires.

3.3.3. Evaluating questionnaires and measures

To assess the potency of our system especially for our proof-of-concept of the CBT scenario and validate our hypotheses, we use questionnaires and objective measures from the literature.

Questionnaires. For the evaluation of the perception of the agent, on a 5-point Likert scale (from 1 (not at all) to 5 (very)), we formulate our questionnaires based on existing questionnaires of human-agent interaction evaluation which are the following:

- Behavior naturalness (Fitrianie et al., 2021; Von der Pütten et al., 2010): e.g. “Is the behavior of the virtual agent artificial?”;
- Behavior human-likeliness (Fitrianie et al., 2021; Von der Pütten et al., 2010): e.g. “Does the virtual agent behave like a human?”;
- Engagement (Fitrianie et al., 2021; Von der Pütten et al., 2010): e.g. "The virtual agent was engaged in the conversation?”;
- Synchrony (Prepin et al., 2013; Louwerse et al., 2012): e.g. “The virtual agent and I were agreeing to each other?”;
- Rapport (Wang and Gratch, 2009; Von der Pütten et al., 2010): e.g. "I think the virtual agent and I established a rapport”.

As CBT is a psychological therapy, to assess the effectiveness of CBT we take the questionnaires from the psychology field which are as follows.

- State-Trait Anxiety Inventory (STAI) (Spielberger et al., 1971): psychological inventory consisting of 40 self-report items for measuring participant’s anxiety level (state and trait with 20 items each scored from 1 (not anxious at all) to 4 (very anxious)), e.g. "I feel nervous”.;
- Kessler Psychological Distress Scale (K6) (Kessler et al., 2002): six-item self-report measure via a 5-point Likert scale (ranging from 0 (none of the time) to 4 (all of the time)) for measuring participant’s psychological distress level, e.g. "How often did you feel so depressed that nothing could cheer you up?”;
- Cognitive Change-Immediate Scale (CC) (Schmidt et al., 2019; Vittorio et al., 2022): five-item self-report measure (rated on a scale from 0 (not at all) to 6 (completely)) that assesses help-seekers’ experience of cognitive change and cognitive skill use during sessions, e.g. "I noticed myself thinking less negatively”. As the CC scale for reporting the user’s experience, the CC is measured once after the session.

The anxiety state change is calculated via STAI-state, which is reported before and after the CBT session, using Eq. (1).

$$Anxiety\,state\,change = STAI\text{-}state_{pre} - STAI\text{-}state_{post} \tag{1}$$

For psychological distress, the participants reported their distress level using the K6 scale before and after the session. Before the experiment, the participants responded to the K6 scale taking into account how they felt over the previous 30 days. After the experiment, they were asked to report whether they felt a change for any of the K6 items after the CBT session. The change in psychological distress level is measured by Eq. (2).

$$Psychological\,distress\,level\,change = K6_{pre} - K6_{post} \tag{2}$$

We also measure the mood change by computing the mood change score, defined in Eq. (3), presented in Shidara et al. (2022). The mood scores ($mood_{pre}$ and $mood_{post}$) are self-reported mood scores (negative mood intensities between 0–100; 0 for happy and 100 for depressed).

$$Mood\,change = \frac{(mood_{pre} - mood_{post})}{mood_{pre}} \tag{3}$$

Measures. To fully evaluate the performance of our system, we also assess it via objective measures. As we focus on the adaptation, it is interesting to investigate whether it was established within the interaction between our agent and the user along with the evaluation of behavior appropriateness. We employ the measures:

- Quantity of movement (ΔQ_{mvt}): quantity of movement (head rotations R_x, R_y, and R_z) defined as in Eq. (4):

$$\Delta Q_{mvt} = \sum_{t=0}^{T} \sqrt{R_{x,t}^2 + R_{y,t}^2 + R_{z,t}^2} \tag{4}$$

  where T is the sequence length.
- Kolmogorov–Smirnov two-sample test (KS test) (Massey, 1951): density probability difference between the agent’s generated behavior and real human behavior to check for the distribution similarities between them, also used in Woo et al. (2023d).
- DTW (Müller, 2007): proximity/resemblance check of the agent’s generated behavior (for one of the experimental conditions of RA, MM, or SP) and user’s behavior against that of the human-human interaction (interactions in the NoXi database Cafaro et al., 2017) to evaluate reciprocal adaptation. To detail, to assess the CBT session, the behaviors of the user and the agent under one of the experimental conditions (RA, MM, or SP) are used to compute the proximity score. This CBT session score is compared with the human-human interaction score which is the average of the scores obtained from the pairs of interlocutors within NoXi. DTW is computed in chunks of 1 min with a stride of 30 s as in Woo et al. (2023d).
- Synchrony (Sync) and Entrainment Loop (EL) (Woo et al., 2023e): synchrony and entrainment loop measures proposed in Woo et al. (2023e) to evaluate the reciprocal adaptation between the agent and the user as in Woo et al. (2023a).

Statistical analysis. A one-sided Welch’s t-test was employed to compare the measuring aspects of user experience and CBT effectiveness for the experimental group pairs of (RA,MM) and (RA,SP) to examine our two hypotheses.

Interpretation. The perception of the agent factors of behavior naturalness, human-likeliness, synchrony, engagement, and rapport are interpreted as rendering a more positive agent impression when the score is above 3.

For the interpretation of the agent’s objective measure, the lower the values are for the metrics of Q_{mvt}, KS test, and DTW, the better the agent performs as it is closer to the human interlocutor within a human-human interaction. Concerning the measures of synchrony and entrainment loop, the higher the value the better they are.

For the measures of CBT, the CBT is effective, showing the improvement of a certain factor, when:

- Mood: mood score (negative mood from pre to post) decreases or high positive mood change score;
- Anxiety: anxiety state (anxiety level from pre to post) decreases or high positive anxiety state change score;
- Psychological distress: psychological distress level (stress level from pre to post) decreases or high positive psychological distress level change score;
- Cognitive change: high CC score (experience of cognitive change and cognitive skill).

3.3.4. Experimental setting

A sample size of 60 French-speaking participants (confidence level of 90 with a margin of error of 10 for the French population with a higher education degree (population portion of 33%)) was chosen for our experiment. We separate them into our 3 experimental conditions of RA, MM, and SP having 20 participants for each condition. Table 2 presents the demographics of the recruited participants per experimental condition (RA, MM, and SP). The inclusion criteria for this study were French-speaking, 18 years or older, and no vision or hearing impairments.

Table 2
Participant demographics per condition (RA, MM, and SP).

		RA	MM	SP
Number of participants		20	20	20
Gender				
	Male	14	11	15
	Female	6	9	5
Age				
	Under 30	17	15	16
	Above 30	3	5	4

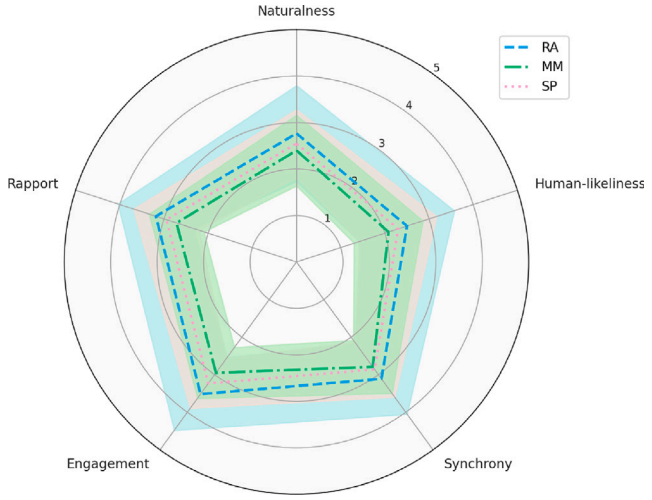


Fig. 4. Perception of the agent's behavior (naturalness, human-likeliness, synchrony, engagement, and rapport). The dotted lines represent the mean value of each condition (RA, MM, and SP) and the colored-filled contour represents the standard deviation of each condition.

4. Results

To validate our hypotheses, we study the impact of our real-time adaptive agent on the perception of the agent, effectiveness for CBT (user mood and state change), and the relation between the agent perception and user mood/state change. For the significance study, we took $p < 0.05$ as the common significance threshold.

4.1. Perception of agent's behavior

We also hold interest in whether our experimental conditions (RA, MM, and SP) influence how the participants perceive the agent (perception of the agent's behavior) along the 5 factors of naturalness, human-likeliness, synchrony, engagement, and rapport.

For the perception of the agent's behavior, we can note, in Fig. 4 and Table 3, that the condition of showing reciprocally adaptive behavior (RA) is perceived as the most natural, human-like, in sync, engaged, and to have a rapport with the participant among the three experimental conditions. An agent showing reciprocally adaptive behavior is indeed perceived more positively along all 5 aspects compared to an agent displaying random behavior (or mismatched behavior; MM) validating our first hypothesis H1 (RA > MM). We also validate that RA condition is better than a still agent (or in a still position; SP) also for all 5 aspects validating our second hypothesis H2 (RA > SP). A significant difference was found between the experimental conditions of RA and MM for the engagement aspect (Mean of RA = 3.52 and SD of RA = 0.964, Mean of SP = 2.95 and SD of SP = 0.678, Cohen's d = 2.15 and p-value = 0.039). No significance was found for the other aspects and for the aspects between RA and SP conditions.

Table 3
Agent behavior perception factors for the conditions of RA, MM, and SP.

(a) Mean and standard deviation.

Factor	RA		MM		SP	
	Mean	SD	Mean	SD	Mean	SD
Naturalness	2.77	1.030	2.40	0.762	2.55	0.728
Human-likeliness	2.50	1.070	2.08	0.779	2.30	0.871
Synchrony	3.11	0.941	2.79	0.706	2.86	0.732
Engagement	3.52	0.964	2.95	0.678	3.23	0.659
Rapport	3.18	0.860	2.72	0.620	2.95	0.743

(b) Significance difference between the condition pairs (RA, MM) and (RA, SP) reported via one-sided Welch's t-test.

Factor	Between RA-MM		Between RA-SP	
	Cohen's d	*P*-value	Cohen's d	*P*-value
Naturalness	1.28	0.210	0.767	0.448
Human-likeliness	1.41	0.169	0.647	0.522
Synchrony	1.22	0.232	0.938	0.355
Engagement	2.15	0.039	1.085	0.286
Rapport	1.97	0.057	0.935	0.356

Table 4
Mean and standard deviation of agent perception objective measures for the conditions of RA and MM. Significance difference between the condition pair (RA, MM) reported via one-sided Welch's t-test.

Measure	RA		MM		Cohen's d	P-value
	Mean	SD	Mean	SD		
ΔQ_{mvt}	12.4	11.8	41.6	31.0	−3.933	<0.001
$KS\ R_x$	0.688	0.274	0.832	0.228	−1.807	0.079
$KS\ R_y$	0.521	0.203	0.705	0.143	−3.324	0.002
$KS\ R_z$	0.652	0.218	0.897	0.136	−4.272	<0.001
$KS\ AU1$	0.413	0.237	0.623	0.165	−3.263	0.003
$KS\ AU2$	0.376	0.224	0.862	0.080	−9.126	<0.001
$KS\ AU4$	0.758	0.241	0.961	0.121	−3.372	0.002
$DTW\ AU1$	115	44.9	128	25.8	−1.105	0.278
$DTW\ AU2$	106	30.3	82.3	26.7	2.604	0.013
$DTW\ AU4$	109	54.8	182	30.8	−5.228	<0.001
$Sync\ AU1$	2.98	3.47	0.00	0.00	3.835	0.001
$Sync\ AU2$	2.33	4.82	0.00	0.00	2.155	0.044
$Sync\ AU4$	15.6	30.4	0.00	0.00	2.293	0.033
$El\ AU1$	173	105	148	88.7	0.817	0.419
$El\ AU2$	222	144	0.00	0.00	6.898	<0.001
$El\ AU4$	365	613	181	170	1.292	0.210

It is interesting to notice that the least performing condition for all 5 factors is the MM condition which shows random and unsynced agent behaviors. It seems that the display of random behavior disregarding the interlocutor rather hinders the impression of the agent. Maintaining a still posture (SP) is preferred to showing non-adaptive behaviors that not considering the participant at all.

We also look at objective measures of DTW, KS test, and synchrony and entrainment loop measures. For the objective study, as we want to see the influence of having adaptive behaviors, we compare our conditions of adaptive (RA) and non-adaptive (MM).

We can check in Figs. 5 and 6 and Table 4 that the condition displaying reciprocally adaptive behavior (RA) performs better than that showing non-adaptive or mismatched behavior (MM) in terms of head movement quantity (via Q_{mvt}), density distribution similarity (via KS test), and DTW resemblance. In addition, we can remark that the adaptive condition (RA) is better than the non-adaptive condition (MM) in being in sync and in entraining its interlocutor's behavior. Via these objective measures, we can also note that an adaptive agent scores better than a non-adaptive one further supporting our prior observation that the RA condition renders a better impression than the MM condition.

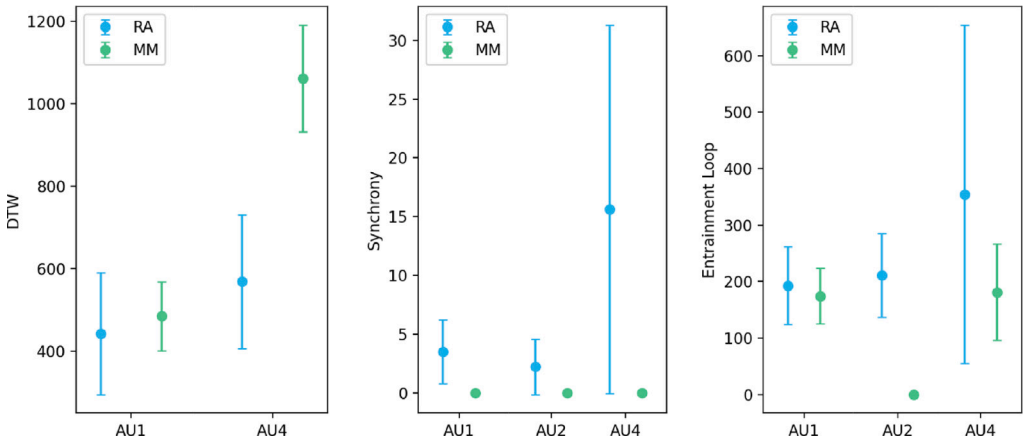


Fig. 5. DTW resemblance, synchrony, and entrainment loop between adaptive (RA) and non-adaptive (MM) conditions.

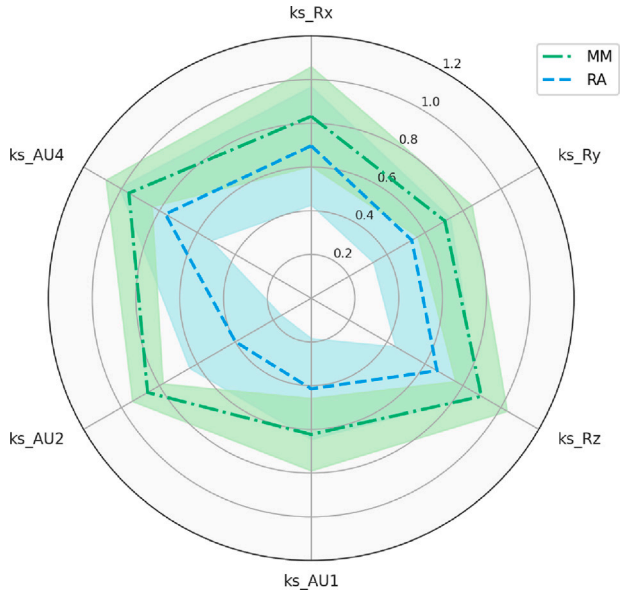


Fig. 6. KS test between adaptive (RA) and non-adaptive (MM) conditions. The dotted lines represent the mean value and the colored-filled contour represents the standard deviation of each condition.

Table 5
Mean and standard deviation of user mood and state measures (pre and post) for the conditions of RA, MM, and SP.

		RA		MM		SP	
		Mean	SD	Mean	SD	Mean	SD
Mood score							
	pre	59.3	24.5	70.0	19.8	51.0	24.1
	post	36.0	23.1	35.6	21.9	33.3	22.1
K6							
	pre	8.05	3.85	6.50	3.00	7.30	2.75
	post	6.10	3.46	5.65	4.72	7.40	3.12
STAI-State							
	pre	33.6	9.20	34.5	11.3	35.6	7.98
	post	32.3	8.84	33.9	10.5	36.8	9.42
STAI-Trait							
	pre	43.5	11.0	42.9	9.86	44.7	7.37
CC							
	post	11.8	7.30	10.2	6.58	11.0	5.20

Table 6
User mood and state measures for the conditions of RA, MM, and SP.

(a) Mean and standard deviation.

Measure	RA		MM		SP	
	Mean	SD	Mean	SD	Mean	SD
ΔMood	0.376	0.251	0.483	0.313	0.246	0.440
ΔSTAI-State	1.40	7.63	−0.50	4.67	−1.20	6.49
ΔK6	1.95	3.39	0.85	2.74	−0.10	2.43
CC	11.8	7.30	10.2	6.58	11.0	5.20

(b) Significance difference between the condition pairs (RA, MM) and (RA, SP) reported via one-sided Welch's t-test.

Measure	Between RA-MM		Between RA-SP	
	Cohen's d	*P*-value	Cohen's d	*P*-value
ΔMood	−1.192	0.241	1.149	0.260
ΔSTAI-State	0.949	0.350	1.161	0.253
ΔK6	1.128	0.267	2.197	0.035
CC	0.705	0.485	0.374	0.711

4.2. User mood and state change

We investigate the global effect of CBT for all three experimental conditions (RA, MM, and SP) combined, with a total number of 60 participants, and the effect of each experimental condition. We assess the effect of the CBT experiment by looking at the factors of mood, anxiety, psychological distress, and cognitive change.

For the factors that represent the change of the participant's state before and after (pre & post) the experiment, we calculate their change. For these factors of change (mood, anxiety, and psychological distress), each participant responds to the corresponding questionnaires (mood score, STAI, and K6) twice (before and after the experiment) to measure the change. As the same questionnaires are used twice, we are in the case of having repeated measures. Thus, we can visualize the difference to check whether there was a change as shown in Fig. 7 and Table 5.

4.2.1. Mood change

Looking at the mood scores before (pre) and after (post) the experiment, depicted in Fig. 7 (left) and Table 5, we can remark that the mood score, which indicates the participant's emotional state of being depressed, decreases. This indicates that CBT treatment using our agent platform helps participants to change their negative mood to a positive one.

When looking at the mood change scores, shown in Table 6, we can remark on the difference in experimental conditions (RA, MM, and SP). Adaptive agent (RA) better ameliorates the participants' mood during the CBT session compared to when the agent retains a still position (SP). Although not statistically significant, the non-adaptive condition (MM)

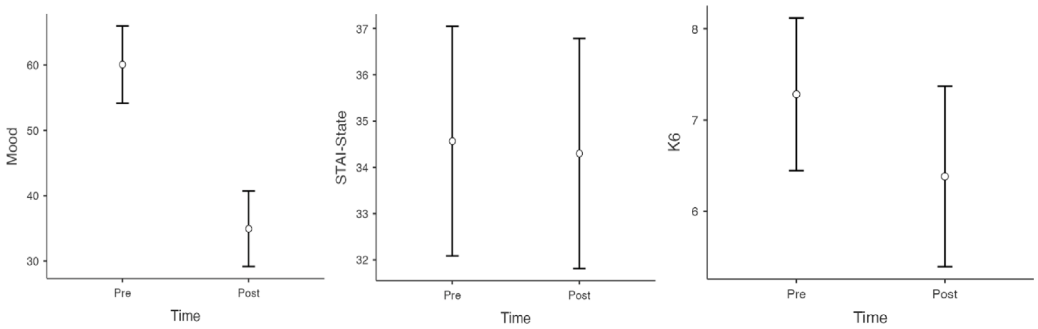


Fig. 7. Change in user mood (mood scores) and states (anxiety level via STAI-State and psychological distress level via K6) before (pre) and after (post) the experiment for all three conditions (RA, MM, and SP) combined.

presents stronger changes in mood scores than the adaptive condition (RA). A potential explanation for this observation is that mood changes correspond more strongly to the feelings associated with situations and automatic thoughts (influenced by the agenda set in the CBT scenario) than other factors such as the adaptability and expressivity of the agent. This may be because mood change asks about the intensity of negative mood when encountering a burdensome situation, as seen in Q3 and Q14 in Fig. 3. Expressivity of agents (MM and RA) was seen to increase the mood change (high positive change) but the user's individual feelings may have had a bigger role in the mood change itself which led the participants in MM to have a higher mood change than those in RA. Thus, to assess general psychological conditions, we also applied the measures of anxiety, distress, and cognitive changes that vary throughout the session. No significance was found between the pairs of (RA, MM) and (RA, SP) for mood change score.

4.2.2. Anxiety state change

For the anxiety level change, we look into the STAI-State inventory in which participants report their current anxiety level.

Before observing the change, we first check the anxiety trait of the different participants via STAI-Trait among the different experimental conditions (RA, MM, and SP). When looking into the STAI-trait score of each condition, found in Table 6, we check that all three levels present similar anxiety traits with no significant differences. This indicates that there are no differences between our three condition groups ($p > 0.1$) which is in line with what we intend as we do not want to reject the null hypothesis (H0).

The anxiety level obtained by STAI-State, in Fig. 7 (center) and Table 5, shows a slight change (decrease in anxiety level for RA and MM, and increase in anxiety level for SP) after the experiment (post) compared to that reported before (pre) when looking at the global change for all three conditions (RA, MM, and SP). This may be interpreted as our CBT agent platform also helps the participants to ameliorate their anxiety levels.

The difference in experimental conditions (RA, MM, and SP) shows different effects on the anxiety level change as seen in Table 6. Showing reciprocally adaptive agent behavior (RA) seems to help in improving the anxiety state. However, agents with no motion (SP) or showing random ones (MM) may deter the state of the participants by increasing their anxiety level instead of relieving them. No significance was found between the condition pairs for anxiety state change.

4.2.3. Psychological distress level change

By observing the psychological distress level obtained via the K6 scale, shown in Fig. 7 (right) and Table 5, we can see that participants' stress level improves after the experiment (post) compared to that reported before (pre), shown by the decrease in stress levels. This may be interpreted as our CBT agent platform is indeed helpful in ameliorating participants' psychological distress.

When looking at the distress level change, shown in Table 6, we can check the difference in experimental conditions (RA, MM, and SP). The psychological distress state seems to improve when the agent is expressive (MM and RA). We can also assume that showing reciprocally adaptive agent behaviors (RA) is more effective in amending the stress level than showing unsynced and random behaviors (MM). When the agent is in a still position (SP), the agent may render the participants in a more stressful state.

Welch's t-test yielded significant variation between the pair of (RA, SP) for distress change (Mean of RA = 1.95 and SD of RA = 3.39, Mean of SP = −0.10 and SD of SP = 2.43, Cohen's d = 2.197 and p-value = 0.035) while no significance was found for the pair of (RA, MM). This result indicates that the stress is relieved with adaptive agents while agents in a still position may rather worsen the stress level of users.

4.2.4. Cognitive change

With the CC scale, results reported in Table 6, we can check that the participants all felt the experience of cognitive change and cognitive skill use during the CBT experiment with our agent platform (CC>0). We can also remark that the condition of RA (Mean = 11.8 and SD = 7.30) performs better for cognitive change compared to SP (Mean = 11.0 and SD = 5.20) and MM (Mean = 10.2 and SD = 6.58) conditions. It seems that displaying reciprocally adaptive agent behaviors (RA) helps in experiencing cognitive change and eliciting cognitive skills. We can also check that showing random agent behaviors (MM) rather leads to a lower cognitive change effect compared to an inexpressive agent (SP). We report no significant statistical differences between the experimental condition pairs.

With this finding, we can validate our hypotheses H1 and H2, showing the relationship of RA>MM and RA>SP respectively, for the CBT effectiveness measures of change in mood, anxiety state, and psychological distress level. Mood change only validates H2 of RA>SP.

4.3. Relation between perception of agent and user mood and state change

Looking at the scores given by the participants, for all three conditions (RA, MM, and SP) combined, along the factors of the perception of the agent, Fig. 8, we can note that the 5 factors are heavily correlated (direct relation) to one another with significant differences ($p < 0.001$). We can remark that the factors of agent perception are closely linked with each other. From Fig. 9, positive correlations between user mood and state change factors are also noticeable, notably between the pairs of (anxiety state change, CC) and (anxiety state change, psychological distress level change). This implies that the mood and state change well compliments each other for the participants' CBT performance evaluation.

We are also interested in looking at the relationship between CBT effectiveness (change in mood, anxiety state, distress level, and cognitive state) and the participant's perception of the agent along these 5 factors (naturalness, human-likeliness, synchrony, engagement, and rapport). We can remark, via Fig. 10, that the anxiety level and the 5 factors of the agent's impression are correlated (STAI-State has a direct

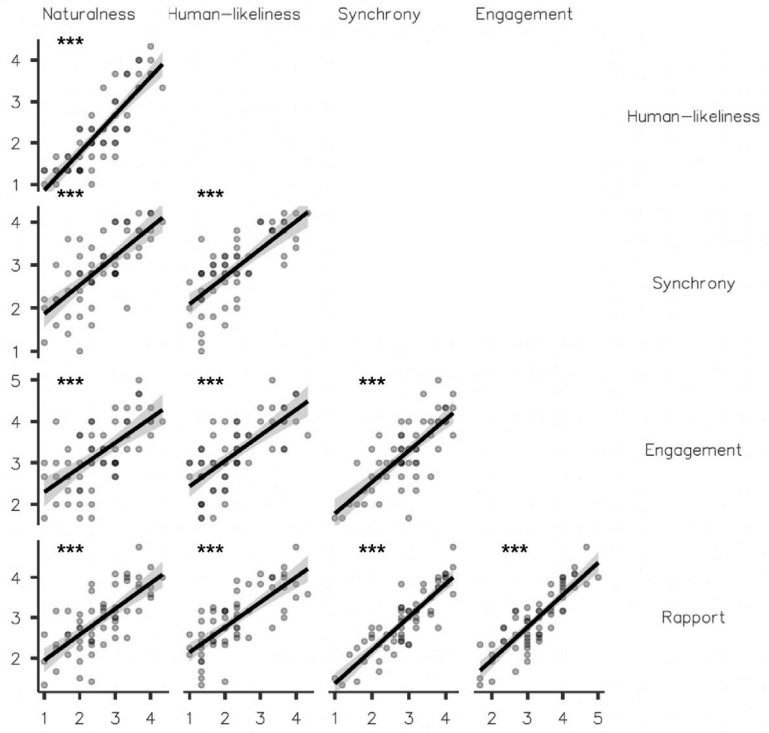


Fig. 8. Correlation between factors of agent perception for all three conditions (RA, MM, and SP) combined with p-values marked as ***: $p < 0.001$.

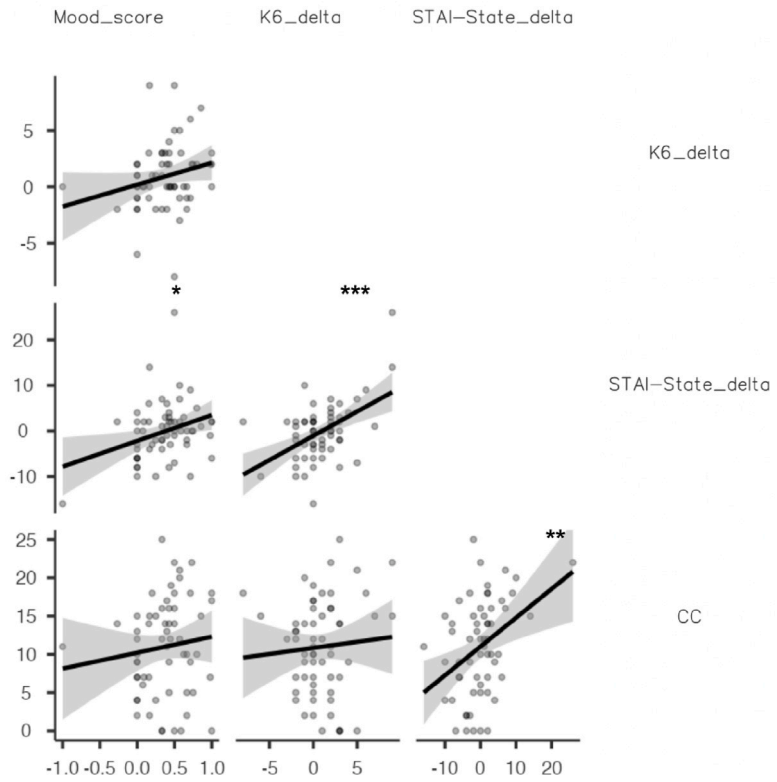


Fig. 9. Correlation between factors of user mood change and state change for all three conditions (RA, MM, and SP) combined with p-values marked as *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.

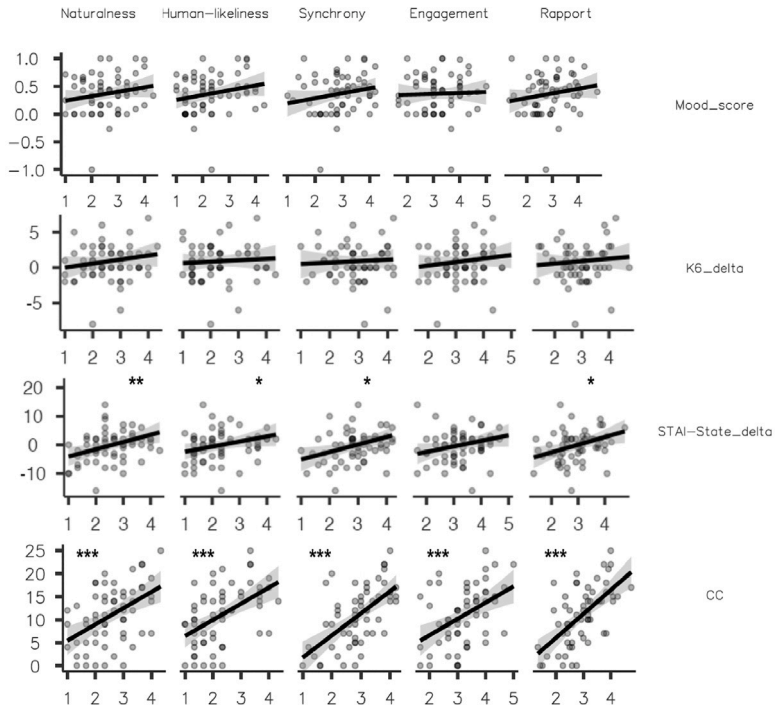


Fig. 10. Correlation between factors of agent perception, user mood change, and user state change for all three conditions (RA, MM, and SP) combined with p-values marked as *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.

relation with the 5 factors) with significant differences for naturalness ($p = 0.008$), human-likeliness ($p = 0.049$), synchrony ($p = 0.01$), and rapport ($p = 0.013$). We can observe the same improvement in cognitive change (CC has a direct relation with the 5 factors) differing significantly for all 5 factors at $p < 0.001$. For the other two factors of CBT effectiveness (mood and psychological distress), no significant correlation was found with any of the 5 factors of the perception of the agent. Our results indicate that the impression of virtual agents is positively correlated to user anxiety state and cognitive change.

5. Discussion

5.1. Result summary

Through subjective and objective evaluations, we were able to validate that the display of adaptive behavior leads to the agent's positive perception of being considered as more natural, human-like, in sync, engaged, and having built a rapport with the participant. The results are in line with previous studies which demonstrated that agents displaying adaptive behaviors increase the user's engagement (Schroder et al., 2011; Ritschel et al., 2017; Weber et al., 2018; Biancardi et al., 2021; Boustani et al., 2021; Woo et al., 2023d), rapport (Huang et al., 2010; Raffard et al., 2018; Amini et al., 2021), interaction synchrony (Raffard et al., 2018; Woo et al., 2023d), and impression of the agent (Bailenson and Yee, 2005; Huang et al., 2010; Biancardi et al., 2021; Woo et al., 2023d). We were able to also remark that a non-adaptive agent is perceived less well than a static one (in a still position). This implies that non-adaptive behavior can rather hinder the impression of the agent. Showing just expressive behavior that does not take into account the interlocutor is worse than not showing any behavior.

Concerning the effectiveness of CBT, we were able to check that our virtual agent system, disregarding the different conditions (RA, MM, and SP), helps deliver CBT in improving users' moods and states of anxiety, psychological distress, and cognitive change. This highlights the advantage of deploying virtual agents in general notably in healthcare. The adaptive condition showed its usefulness in having higher effects of CBT. We noticed that agents showing random expressive behavior can still improve the impact of the application in terms of mood change, stress, and cognitive change. These results are consistent with findings in the medical and psychological fields that rapport (VandenBos, 2007) affects the effectiveness of CBT (Asay and Lambert, 1999; Ardito and Rabellino, 2011; Norcross and Lambert, 2018), and in the engineering field that nonverbal behavior affects rapport (DeVault et al., 2014; Huang et al., 2010; Raffard et al., 2018) as defined in the context of human-agent interaction (Gratch and Lucas, 2021). This may be related to the presence of the agent that the participants feel during the CBT session. Expressive agents may heighten the feeling of the agent being physically present with the participants and thus leading to a psychological ease to interact and benefit the effect of the interaction and thus the CBT treatment.

Nevertheless, looking at the anxiety state change, we can remark that agents displaying random behaviors (MM) or remaining still (SP) may worsen the anxiety level of the users. Thus, we need to be careful when implementing agents for medical therapy as such agents (MM and SP) may significantly raise the user's anxiety levels.

We also found correlations between the factors of perception of the agent's impression, between the user's mood and states, and between the perception of the agent's impression and the user's anxiety state and cognitive change. The positive correlation observed between the factors of perception of the agent's impression, revealing a halo effect, shows that the factors are indeed tightly connected which can be grouped as a global impression of the agent. The correlations found between user mood and state change factors, especially the positive relation between anxiety and distress state changes, denote that these factors complement one another in showing the effect of CBT. The interdependence seen between agent perception and user states (anxiety and cognitive) may be interpreted as the user states can be ameliorated by improving the agent's impression which can be done by endowing the agent with the skill to adapt.

5.2. Limitations and future work

Our real-time adaptive virtual agent system proved its usefulness for the proof-of-concept of CBT. To manifest that our system is applicable

to various domains, its use needs to be confirmed with other use cases. Notably, for other agent-based health interventions, similar adaptive agents' nonverbal behaviors could be applied. To address this, we are currently applying our system to Social Skills Training (SST) (Bellack et al., 2013), a behavioral therapy for improving social skills in people, to demonstrate our system's applicability and effectiveness to other usages. The real-time adaptive SST agent system has been implemented (Saga et al., 2023) and like CBT, we expect to further improve the SST performance with our adaptive agent.

We also envision conducting a cultural comparison of the impact of adaptive virtual agents for the same application of CBT. People from cultures behave differently according to their cultural style. The way of adapting their behavior may thus vary across cultures and the adaptation may be perceived and interpreted in another way. We want to investigate whether the adaptive virtual agent presents such cultural-dependant behavioral differences and whether it can also enhance the user experience and CBT performance for the population of another culture in the same manner. An interesting vision is to check if the adaptation mechanisms can be generalized across cultures. This could be done by trying out our system developed for a specific culture to the population of another culture.

For this study, we tested our agent with healthy participants as CBT is a treatment that is also useful for the general population. It could be also tested on participants with mental health issues such as patients with Autism Spectrum Disorders (ASDs) to check the effectiveness on real patients.

In addition to producing agent's adaptive behaviors of facial expression and head/gaze movements, we plan to extend the adaptive behavior generation by modeling and integrating a real-time full body agent adaptive behavior generation model (notably for upper body gestures including hand gestures and torso movements) to our system. Other forms of adaptation could also be modelized such as the adaptation to users' personality traits and communication strategies.

A limitation of our system at the current stage can be its physical portability. For the system to function in real time (at 25 Hz), heavy computation is required to be done in the background. The dependency on our physical devices (computers with high performance) is inevitable for our system. This means that our virtual agent is not yet deployable for lightweight devices and thus cannot be used everywhere easily such as on mobile devices. We intend to try to respond to this limitation by better optimizing the computation while assuring the real-time functioning of our system.

Also, the fact that no real corpus of CBT interactions is available is another obstacle that led us to use a natural conversation-based NoXi database. For a more adaptive agent for CBT, we need to collect and use a new corpus of real patient-therapist CBT sessions recorded with front-facing cameras.

6. Conclusion

Adapting to other people is an essential communication skill. Virtual agents interacting with their users need to know how to adapt to their interlocutors to provide a lively and interesting interaction. In this work, we developed a virtual agent endowed with adaptation capacity that is capable of functioning in real time. The agent renders continuous verbal and nonverbal behaviors at the frame-level. The healthcare application of CBT was selected as a proof-of-concept to demonstrate the use of our adaptive agent. The agent showed its utility in delivering CBT by enhancing users' moods and states (anxiety, psychological distress, and cognitive). Showing adaptive behavior can further improve the user experience in terms of the user perception of the agent's impression along the factors of naturalness, human-likeliness, synchrony, engagement, and rapport. However, expressive agents are not always positively perceived since non-adaptive ones may rather deter the user's impression. Inexpressive agents that remain in a still posture also risk worsening the CBT effect by heightening the user's anxiety level. Therefore, we should be careful when designing an agent for the public, especially for the elderly or sensitive populations. In addition, a new human-agent interaction (HAI) database, HAI-CBT database, has been collected. The database contains CBT interaction video recordings, extracted OpenFace and openSMILE data, transcripts, and users' questionnaire responses. As there is a lack of HAI database that is publicly available, this new database will enable the HAI research community to advance further. Adaptive agents are full of promise as they can ameliorate user impressions and be employed for various applications.

Ethics

The study involving human participants was reviewed and approved by Sorbonne University (reference number: CER-2023-WOO-Test Perceptif). The participants provided their written consent to participate in the experiment of this study and to give the authorization of use concerning the collected data.

CRediT authorship contribution statement

Jieyeon Woo: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kazuhiro Shidara:** Methodology, Software, Writing – review & editing. **Catherine Achard:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Hiroki Tanaka:** Writing – review & editing. **Satoshi Nakamura:** Writing – review & editing. **Catherine Pelachaud:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jieyeon Woo reports financial support was provided by ANR-JST-CREST TAPAS Project. Jieyeon Woo reports financial support was provided by IA ANR-DFG-JST Panorama Project. co-author is an associate editor of IJHCS -C.P.

Data availability

Data is available on request after signing the EULA form. To get the HAI-CBT database please contact Catherine Pelachaud[9].

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